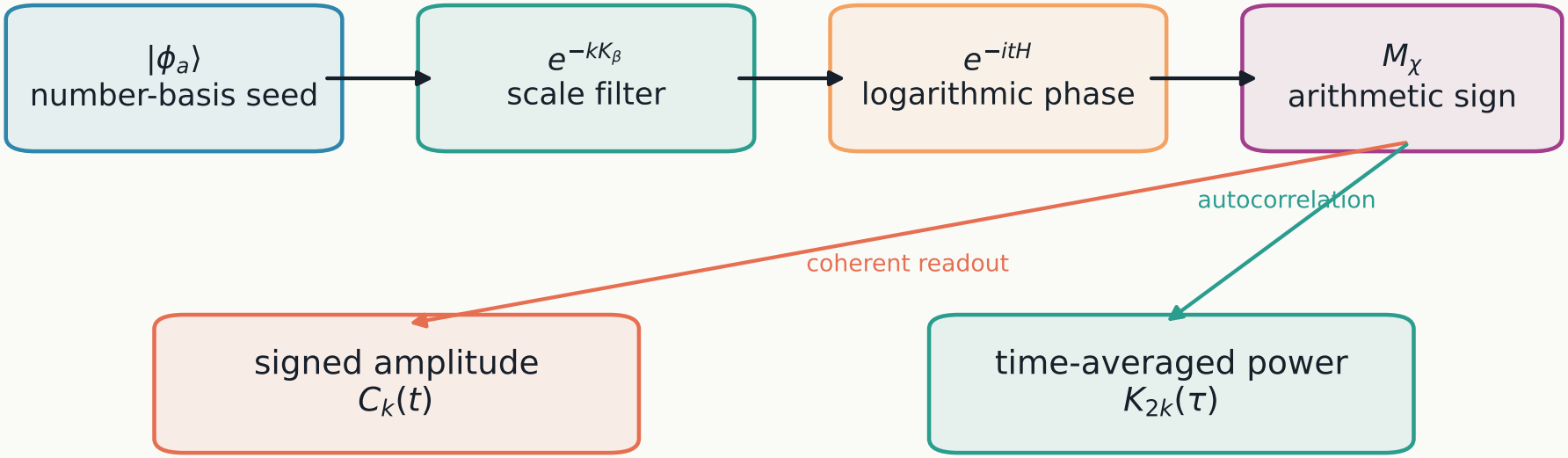Number-basis realization of the signed recurrence process